

managing memory and data input-output from multiple components and peripherals. All that task, simultaneously with each clock tick is no minor feat. SoCs do not implement that many instruction sets and their control system is not designed for such workloads.

2. Since CPUs can get lots of work done using different instruction sets as well as higher frequencies, they are used to build super-computers and servers.
3. If you have tiny amounts of RAM, it would limit the work the CPU can do. Systems low on memory cannot work on large data sets at fast speeds because suspending data to disk and bringing them back is a non-trivial wastage of time and processing power. Although SoCs contain good amounts of RAM, when you want to create a system that can address terabytes of RAM, you turn to the CPU (<http://www.dell.com/in/business/p/poweredge-r930/pd>). Being able to host and address terabytes of RAM is a luxury SoCs cannot have.
4. CPU is portable in implementation: You can upgrade the system by upgrading to a better CPU without tearing down the rest of the system. Or, you can add or replace RAM and the CPU would work fine.

When both fail

SoCs are full systems and CPUs are beasts around which full systems can be built but they are not ideal for all use-cases.

For example, what would a microwave oven use for controlling heat and timing? A CPU, or a SoC? Probably neither. A simple microwave controller would need a ASIC chip (https://en.wikipedia.org/wiki/Application-specific_integrated_circuit) because that would be sufficient to display small amounts of information on a LED screen, run a timer and sound the beeper. Both CPU with its processing power and SoC with myriad functionality would be an overkill. If, however, it was to be a IoT device, an SoC would probably make sense. A car's dashboard system can do without an SoC too unless it is a self-driving car which needs an array of sensing tools and algorithms to drive through traffic. Also, they need GPUs to handle compute tasks.

There are very few computational requirements in today's world which do not (or cannot) use either a SoC or a CPU. Such requirements are either too small or too large and complex to be fulfilled solely by either of the two types. While larger requirements can be solved by large computing clusters with different types of processors (DSPs, GPGPUs and CPUs), smaller ones can be taken care of by ASIC implementations. **d**